Practice with the Low Power and Longue Range Network LoRa/LoRaWAN

Presentation of the LoRa/LoRaWAN long-distance network

It is a network designed for connected objects that are isolated or far from the network infrastructure by several kilometers (1 to 40km). The transmitted signal is easy to understand when received in order to be able to cover long distances. On the other hand, its bitrate is limited to a few Kilobits per second. The signal shape can be configured to focus on distance, robustness, operating time, bitrate or safety.

Comparative table of LoRa/LoRaWAN technology compared to Sigfox, LTE (cellular network)

	LoRaWAN	Sigfox	NB-LTE
Modulation	Spectrum Spreading (CSS)	Ultra Narrow Band, GFSK, BPSK	
Frequency	169, 433, 868MHz	868MHz	
Channel Width	125,250, 500kHz	100Hz	200kHz
bitrate	290bps to 50Kbps	10 to 1000 bps	20Kbps
Packet Size	Up to 242 bytes max	8 (DL data) to 12 (UL data) bytes	
Transmission time of 1 packet	~ 1 second	~ 2 seconds	
# message per day	unlimited (duty cycle 1%)	140/day	unlimited (duty cycle)
Correction Code	Code Rate 1 à 4		
Immunity to interference	Very High	Low	Low
Transmitting Power	+20dBm (100mW)	+20dBm (100mW)	+20dBm (100mW)
Link budget (Bilan de liaison)	154dB SF12 (-134dBm) SF8 (-124dBm)	151dB	150dB
Receiving distance	30km (in view LoS)	20km (in view LoS)	
Energy Efficiency	Very High	Very High	Average/High
Consumption	12uA sleep 13mA RX (11mcu+2rx) 30mA TX (14dBm) 110mA TX (20dBm) 400mA TX (23dBm) 700mA TX (30dBm)	61mA TX (20dBm)	
Battery Life 2000mAh	105 months (>8 years)	90 months (>7 years)	2 years ?
Security	AES128	No	No
Operator in France	Orange Bouygues (4300 antennas, 86 % of France)	SFR (1500 antennas 91 % of people	15000 antennas 3G/4G

Semtech company, located in Grenoble, France, created this communication technology. There are currently 2 companies marketing this product: Semtech SX1272/73 and Hoperf RFM95/98

The component used for the practices is:
- ARM Cortex-M0 microcontroller + LoRa module: https://www.adafruit.com/product/3178

In order to do all the exercice, I ask you to send me at the end of this practice, a file with one screenshot of your screen for each exercice.

First Exercise on your own computer: Compile « blink » program

Install Arduino IDE 1.8.XXX

Then, installation of the plug-in to compile on the target architectures: adafruit ARM-M0

To install the SAMD compilator for the ARM-M0 processor

Run Arduino tool then File → Preferences
https://adafruit.github.io/arduino-board-index/package_adafruit_index.json
(If there is already content, the URLs must be separated by a comma).

Tools → board type → board management
Install « Arduino SAMD boards (32-bit ARM cortex-M0+) »
then, install « Adafruit SAMD boards »

Choose the target hardware Tool → Board → « adafruit feather M0 express »

Test the blink example, to validate the correct installation of the compiler (File → Example)
Because you do not have any microcontroler connected on your own computer you can go up to compile the binary file but you cannot download it.

Take a screenshot of your compilation with the successful compilation message on it and put it in your document file.

Second practice: Running a « Hello World » program on a remote classroom computer

Run the Arduino tool, select the target hardware « Adafruit Feather M0 Express »

Select the serial port COM3 (it can be COM4 or COM5 according to your OS) : tools → port → COM3
and open the Serial Window to monitor the output of your microcontroler, tools → Serial Monitor

Create a new program file and add the follow line in the setup() function, which is run at the launch step.

Serial.begin(115200) ;

This line will configure the COM3 virtual serial port at 115200 bps.

Add the following lines in the loop() function, which is executed each time it ends
Serial.println(Hello World) ;   // This will add the message on the Serial Window at the running step
delay(3000) ;   // This will add a 3-second delay between each message

Then compile and upload this program to test the microcontroler : sketch → upload

Take a screenshot of your screen with the serial window on it and put it in your document file.

Third practice: Receiving a LoRa signal from a component in the room

(This step has been already done for you). Install the Radiohead library to use the LoRa/LoRaWAN communication modules by inserting the file .zip (croquis→include a library→ zip file)
http://www.airspayce.com/mikem/arduino/RadioHead/
http://www.airspayce.com/mikem/arduino/RadioHead/RadioHead-1.105.zip

Creating the reception signal program

```
#include <SPI.h> // The LoRa module includes an interface with SPI bus
#include <RH_RF95.h> // Radiohead Library for LoRa communications
```

```
#define RFM95_CS 8
#define RFM95_RST 4
#define RFM95_INT 3

RH_RF95 rf95(RFM95_CS, RFM95_INT);

uint8_t buf[255];
uint8_t len = sizeof(buf);

void setup() {
  Serial.begin(115200);

  rf95.init();
 rf95.setModemConfig(RH_RF95::Bw125Cr45Sf128);
                                                          // Bandwidth=125kHz Code rate 4/5 Spreading factor =2^7
  rf95.setFrequency(868.1); // 868.1, 868.2 or 868.3
}

void loop() {
  if (rf95.waitAvailableTimeout(4000)) { // Wait for a packet for 4 seconds
    if (rf95.recv(buf, &len)) {
      Serial.println((char*)buf);
      Serial.print("RSSI: ");
      Serial.println(rf95.lastRssi(), DEC);
      Serial.print("SNR: ");
      Serial.println(rf95.lastSNR(), DEC);
    }
  }
  else Serial.println("No reception");
}
```

Take a screenshot of your screen with the serial window on it with succesful reception and put it in your document file.

Forth practice: transmit a LoRa signal with a specific spreading factor

In the setup function, add the configuration of the transmit power :

```
rf95.setTxPower(13, false); // +13dBm
```

In order to send periodically a packet each 2 seconds

```
void loop() {
  message="LO53 Practice #";
  message+=i;
  i++;
  message.toCharArray((char*)buf, message.length()+1);
  buf[message.length()+1] = 0;
  rf95.send((uint8_t *)buf, message.length()+2);
  rf95.waitPacketSent();
  Serial.print("Sent Message:");
  Serial.println(message);
```

```
  delay(2000);
}
```

Add a LED blink for each transmission, in order to visualise the time to send.

Take a screenshot of your screen with the serial window on it with succesful reception and put it in your document file.

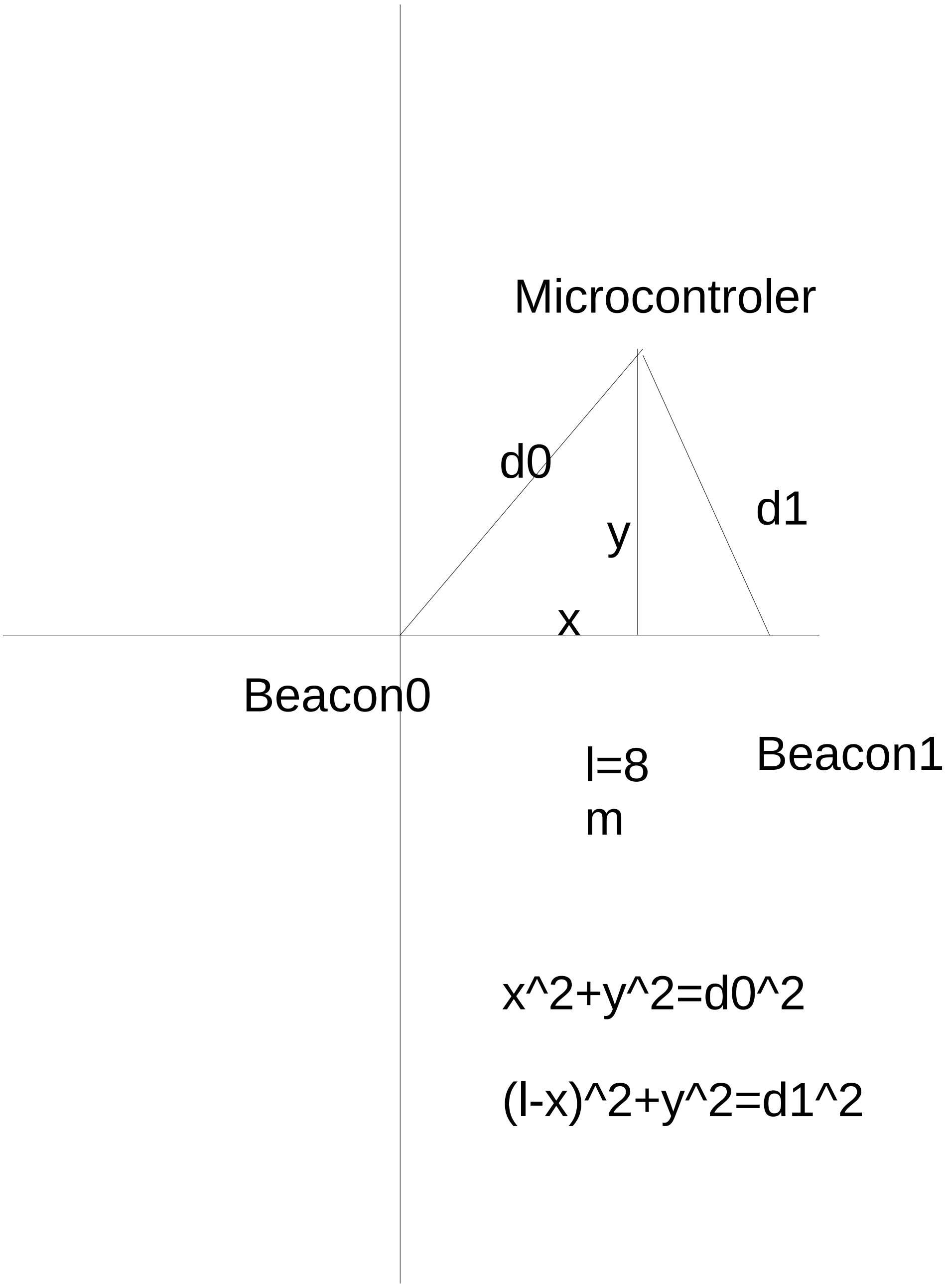
Fifth Practice: Estimate of the pseudo-distance between sender and receiver using Signal Strength.

3 LoRa componants have been installed in the room at various locations. Each componant is sending a periodic message with a Beacon ID and a location.
According to the location of your componant, you will receive the Beacons with a specific RSSI (Received Signal Strength Indication).
Write a program which receive a message, get the corresponding RSSI level signal and calculate the pseudo-distance using the Friis formula. You have to print a message on the serial window such as :
ID=0, RSSI=-60dBm, distance=5.0m
ID=1, RSSI=-70dBm, distance=6.0m
ID=2, RSSI=-80dBm, distance=7.0m
Take a screenshot of your screen with the serial window on it with succesful reception and put it in your document file.

6th Practice: Estimate the coordinates of your component in the classroom.

In the messages of the beacon, you can read the coordinates which are (0,0) for Beacon0, (8,0) for Beacon1 and (8,0) for Beacon2. These 3 beacons are drawing a right triangle. Using the beacons coordinates and the pseudo-distances, please calculate the 2D-coordinates of your component.

Take, 0<x<8 and 0<y<8, If d0<8 and d1<8 and d2<8, x-axis and y-axis will be positive numbers



$$x^2+y^2=d0^2$$

$$(l-x)^2+y^2=d1^2$$